

Three Severe Tests of a Multi-Agent Model of Paradigm Change

An experiment-design document at the intersection of philosophy of science and computational social science

Forest-walk synthesis
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Abstract

We design three experiments on a multi-agent active-inference model of scientific paradigm change. The model exists in two interchangeable implementations: a *structural* Gaussian information-form Bayes-net in which a paradigm *is* a precision matrix (`src/structural/`), and a *categorical epistemic POMDP* in which a fixed true paradigm θ^* is read through theory-laden likelihoods and probed by experiments chosen to minimise expected free energy (`src/pomdp/`). The model’s central empirical claim to date is that its dynamics are *monostable*—a single attractor at the truth—and that apparent “lock-in” is *self-censorship* (a cost-driven refusal to run discriminating experiments) and *paradigm-governed precision silencing*, not genuine bistability. Following Mayo’s principle that a claim is only established when it has survived a test it would probably have failed if false, each experiment is built around an explicit falsifier. Experiment 1 probes the *ontology* (is there a second basin?) via rate-resolved hysteresis. Experiment 2 probes the *individual epistemics* (when does a research programme seal itself?) by mapping the progressive/degenerating frontier in the cost $\times$ conviction plane. Experiment 3 probes the *collective epistemics* (when can a structured society break the seal?) via committed-minority tipping under varying trust topology. All manipulated variables are named code parameters with ranges; all outcomes are existing order parameters.

Contents

1 The model under test

1.1 Two implementations of one object

(S) Structural Bayes-net. Each agent holds a Gaussian belief over d interdependent commitments in information form $(\mathbf{\Pi}, \mathbf{h})$ with $\mathbf{\Pi} = \Sigma^{-1}$ and $\mathbf{h} = \mathbf{\Pi}\boldsymbol{\mu}$. A *paradigm* is the off-diagonal structure of $\mathbf{\Pi}$: a dense hub block (the Lakatosian *hard core*) coupled to peripheral commitments (the *protective belt*). The canonical scenario is the $d=10$ phlogiston network (`src/structural/phlogiston.py`) with three DISAGREEMENT_NODES on which phlogiston and oxygen priors conflict. Per round the population does

$$\underbrace{\mathbf{\Pi}_f = \sum_j W_{ij} \mathbf{\Pi}_j, \quad \mathbf{h}_f = W \mathbf{h}}_{\text{FUSE (trust-weighted pooling)}} \longrightarrow \underbrace{\mathbf{\Pi} = \mathbf{\Pi}_f + \sum_k \rho_k \mathbf{H}_k^\top \mathbf{H}_k / \sigma_o^2}_{\text{OBSERVE (attention-gated Fisher deposit)}}, \quad (1)$$

where the per-channel *evidential precision* ρ_k is the model’s attention. Bayesian Model Reduction (BMR) prices the removal of a node by the closed-form log Bayes factor ΔF (Savage–Dickey),

and the *carry-over* $\Pi_{ab}\Pi_{bb}^{-1}\Pi_{ba}$ (Schur complement) is the literal cost of revising the core. The order parameter is

$$m_t = \frac{\bar{\mu}_{\text{disagree}}(t) - \mu_{\text{phlog}}}{\mu_{\text{oxy}} - \mu_{\text{phlog}}} \in [0, 1], \quad m_t=0 \text{ phlogiston held, } m_t=1 \text{ oxygen held.} \quad (2)$$

(P) Categorical epistemic POMDP. A fixed true paradigm $\theta^* \in \{0, \dots, K-1\}$ generates outcomes through a theory-laden likelihood $A_{\text{world}}[o, \theta, a]$ (columns differ across θ : the same outcome is read differently under different theories). Each agent infers $q(\theta)$ by exact categorical Bayes from a chosen experiment’s outcome and a trust-routed social report $o_{\text{social}} = Tq$ with channel fidelity `q_reliability`. Experiments a are chosen by softmax over the negative expected free energy

$$-\text{EFE}(a) = \underbrace{\mathbb{E}_o \text{KL}[q(\theta | o, a) \| q(\theta)]}_{\text{epistemic (salience)}} + \underbrace{\mathbb{E}_o C[o]}_{\text{pragmatic}} + \beta_U \text{bu}(a) - \underbrace{\text{cost_scale} \cdot c[a]}_{\text{experiment effort}}, \quad (3)$$

with policy $q(a) = \text{softmax}(\text{gamma_policy} \cdot (-\text{EFE}))$. Order parameters returned by `run_fast`: `mean_qB` (mean belief in the truth), `occ_B` (fraction with MAP= θ^*), `p_discrim` (policy mass on the most-discriminating experiment), `chosen_d` (mean discriminability of chosen experiments).

1.2 The claim being probed, and the methodological stance

The standing empirical claim (notebooks 18–20) is:

The model is monostable—every trajectory relaxes to $\mu = \theta^$. “Lock-in” is not a second attractor; it is (i) cost-driven self-censorship, the EFE-rational refusal of a confident agent to run an expensive, low-information refuting experiment, and (ii) paradigm-governed precision silencing via the **core_governance** knob. Trust-weighted social disagreement can break the self-seal.*

We adopt Mayo’s *severe testing* criterion [?]: a claim is corroborated only by a procedure that *probably would have produced a result discordant with the claim, were the claim false*. Each experiment therefore foregrounds the specific result that, if observed, refutes the standing claim. The three experiments are ordered by level of description—ontology, individual, collective—mirroring the Lakatosian distinction between a programme’s hard core, its problemshifts, and its competition with rivals [?].

2 Experiment 1 — Rate-resolved hysteresis: is the object bistable?

2.1 Motivation

Kuhn [?] describes paradigm change as path-dependent and gestalt-like: once a community converts, returning the evidence to its earlier state does not return the community. That is the phenomenology of *bistability* (two attractors with a separating basin boundary). The model claims the opposite: *monostability with tracking lag*. The two are observationally identical under a one-directional drive, which is exactly why a severe test is needed. The discriminating instrument is a *reversible* drive applied at several speeds: a tracking lag yields a hysteresis loop whose area vanishes as the drive slows, whereas a genuine fold leaves a residual loop in the quasi-static limit.

2.2 Design

We drive the population with a triangular evidence schedule. In (S) we ramp the world commitment ϕ_{true} on the disagreement nodes from the phlogiston pole to the oxygen pole and back,

replacing the single `t_shift` flip by a controlled ramp of rate r (pole-to-pole in $1/r$ steps). In (P) we ramp the prior/evidence balance equivalently by scheduling which paradigm the environment samples from. For each ramp we record the loop in (evidence, m_t) space and compute the loop area

$$A_{\text{hys}}(r, g) = \oint m_t d(\text{evidence}), \quad (4)$$

as a function of ramp rate r and the lock-in knob (`core_governance` g in (S); `cost_scale` and `gamma_policy` in (P)). We additionally run a *basin probe*: initialise at the oxygen pole, apply a transient phlogiston evidence pulse of amplitude Δ and duration τ , and record whether the population returns to oxygen or is captured.

2.3 Manipulated variables

Parameter	Implementation	Levels
ramp rate r	both	$\{1/10, 1/30, 1/100, 1/300\}$ (steps $^{-1}$); quasi-static = $1/300$
<code>core_governance</code> g	(S)	$\{0, 1, 10, 100, 1000\}$
<code>cost_scale</code>	(P)	$\{0, 0.25, 0.5, 1.0\}$
<code>gamma_policy</code>	(P)	$\{1, 5, \infty\}$ ($\infty \equiv \text{greedy}=\text{True}$)
pulse amplitude Δ	both	5 levels spanning $[0, 2(\mu_{\text{oxy}} - \mu_{\text{phlog}})]$
pulse duration τ	both	$\{2, 5, 10, 20\}$ steps

2.4 Fixed parameters

$d=10$, $N=60$, `sigma_o`=1.0, `base_prec`=1.0, `hub_coupling`=0.8, `mass_prec`=2.5 (S); $K=2$, $\theta^*=1$, `x_grid`=(0.1, 0.5, 1.0, 2.0, 3.0), `q_reliability`=0.80 (P). $n_{\text{seeds}}=40$. Network: Watts–Strogatz, `mean_degree`=4.

2.5 Outcomes and analysis

Primary outcome: the slope $\partial A_{\text{hys}}/\partial r$ near $r \rightarrow 0$ and the intercept $A_{\text{hys}}(0, g) \equiv \lim_{r \rightarrow 0} A_{\text{hys}}$ obtained by linear (and saturating) extrapolation across the four ramp rates. Secondary: fraction of seeds captured at horizon in the basin probe, as a function of (Δ, τ) , giving an empirical basin boundary; escape-time distribution in (P) under finite vs. infinite `gamma_policy`.

Hypothesis 1 (Monostable, standing claim). $A_{\text{hys}}(0, g) = 0$ for all finite g and all finite `gamma_policy`: the loop area extrapolates to zero in the quasi-static limit; the basin probe always returns to oxygen for finite τ .

Hypothesis 2 (Bistable, Kuhnian alternative). $\exists g^*$ such that $A_{\text{hys}}(0, g) > 0$ for $g > g^*$: a residual, rate-independent loop survives; the basin probe exhibits a sharp (Δ, τ) capture boundary.

Prediction 1. From notebooks 19–20 we expect Hypothesis ?? to hold for all finite settings, with $A_{\text{hys}} \propto r$ (pure tracking lag), and a degenerate fold appearing only in the singular limit `gamma_policy`= ∞ (`greedy=True`), where escape probability is exactly zero. `core_governance` should rescale the relaxation rate (hence the finite- r loop area) without creating a residual loop.

Severe test / falsifier. If the loop area extrapolated to $r \rightarrow 0$ is statistically indistinguishable from zero (one-sided test, $\alpha=0.05$, against the $n_{\text{seeds}}=40$ spread) at every finite g , monostability survives a test it would have failed had a second basin existed. Conversely, a residual loop at any finite g refutes the standing monostability claim. Reporting A_{hys} at a single ramp rate is *not* a severe test and is explicitly disallowed.

3 Experiment 2 — The degenerating-programme frontier: when does the population stop running the crucial experiment?

3.1 Motivation

Lakatos [?] distinguishes *progressive* research programmes (which keep risking and corroborating novel predictions) from *degenerating* ones (which expend their effort on post-hoc self-protection). In this model that distinction is not about belief but about *behaviour*: a degenerating programme is one whose agents stop choosing the discriminating experiment. By the Duhem–Quine logic, the “crucial experiment” is never logically forced; here it is *economically* avoided—for a confident agent the refuting experiment has low expected information gain (Eq. ??, epistemic term) and nonzero cost, so EFE rationally deselects it. The key non-obvious claim (notebook 19) is that this self-censorship appears *even at zero cost* above a conviction threshold. Experiment 2 maps the frontier and gives the zero-cost null its best chance to survive.

3.2 Design

Fully crossed factorial in (P) over experiment cost, initial conviction in the *wrong* paradigm, and policy precision; replicated as a robustness check in (S) via `experiment_bias` and `core_governance`. We track the *behavioural* order parameters `p_discrim` and `chosen_d` over the horizon, alongside the *doxastic* `mean_qB`. The diagnostic of a degenerating programme is the joint event

$$(\text{chosen_d}(t) \text{ declining}) \wedge (\text{mean_qB}(t) < \tfrac{1}{2} \text{ stuck}),$$

i.e. the population both *stops looking* and *stays wrong*.

3.3 Manipulated variables

Parameter	Implementation	Levels
<code>cost_scale</code>	(P)	{0, 0.1, 0.25, 0.5, 1.0, 2.0}
initial conviction D_0 on wrong θ	(P)	{0.5, 0.6, 0.7, 0.8, 0.9, 0.95}
<code>gamma_policy</code>	(P)	{1, 5, ∞ }
<code>cost_kind</code>	(P)	{discriminability, power, vector}
<code>experiment_bias</code>	(S)	{0, 0.25, 0.5, 0.75, 1.0}
<code>core_governance g</code>	(S)	{0, 1, 10, 100}

3.4 Fixed parameters

$K=2$, $\theta^*=1$, `theta_vals`=(0, 1), `beta_U`=0 (switched off here so the effect is purely epistemic+cost; `beta_U` is varied separately in a supplementary sweep), `n_o`=5, `x_grid`=(0.1, 0.5, 1.0, 2.0, 3.0), `q_reliability`=0.80, $N=60$, `horizon` =200, n_{seeds} =40.

3.5 Outcomes and analysis

Per cell we summarise (i) the terminal `chosen_d`, (ii) its time-slope over the second half of the run, and (iii) terminal `mean_qB`. The *self-censorship boundary* is the contour in the (`cost_scale`, D_0) plane where terminal `chosen_d` drops below a fixed fraction (say 0.5) of its cost-zero, low-conviction value. This contour *is* the progressive/degenerating frontier, drawn quantitatively.

Hypothesis 3 (Intrinsic self-censorship). *At `cost_scale`=0 there exists a conviction threshold $D_0^c \approx 0.7$ above which `chosen_d` collapses under `gamma_policy`= ∞ : self-censorship is not purely a cost artifact.*

Hypothesis 4 (Cost-only self-censorship, null). *At `cost_scale=0`, `chosen_d` never collapses for any D_0 or `gamma_policy`: self-censorship requires nonzero experiment cost.*

Prediction 2. *Hypothesis ?? holds under greedy policies; under sampled (`gamma_policy` finite) policies the population almost always eventually runs the crucial experiment and escapes, so the collapse is conviction- and γ -gated, not cost-gated. The frontier should shift toward lower D_0 as `cost_scale` rises. The (S) replication should show `rho_k` on the disagreement channels collapsing along a homologous contour in (`experiment_bias`, g).*

Severe test / falsifier. Hypothesis ?? is the null we are obliged to give a fair hearing: run the complete $D_0 \times \text{gamma_policy}$ sweep *at exactly* `cost_scale=0`. If `chosen_d` stays high everywhere, the “epistemic self-censorship” story is refuted and the mechanism is merely pragmatic cost. Multiple realisability is itself a test: a self-censorship contour that appears in (P) but *not* in the (S) `rho_k` silencing (or vice versa) indicates an implementation artifact rather than a model mechanism, and must be reported as a failure to corroborate.

4 Experiment 3 — Committed-minority tipping: when can a structured society break the self-seal?

4.1 Motivation

The first two experiments treat the population as effectively homogeneous. The computational-social-science question is whether the *topology* of trust decides the outcome. Centola [?, ?] distinguishes *simple* contagion (one exposure suffices; conversion roughly linear in the seed fraction) from *complex* contagion (conversion needs reinforcement from several trusted sources, producing a sharp critical fraction f^* that depends on network structure—famously near 25%, and lowered by wide bridges). Here a committed minority is a *vanguard* that keeps running the crucial experiment and emits the correct report; the legitimate rescue channel is the structural EFE *social* drive `social_weight` β_s , which re-opens a self-sealed channel when a *trusted peer who has actually looked* disagrees (`notes/efe_attention.tex`). Notebook 10 already hints that an *isolated* high-centrality vanguard overshoots and is discounted—a prediction we now test directly.

4.2 Design

Take a mainstream population pre-sealed by the Experiment-2 settings (`cost_scale`>0 or g >0 above the self-censorship frontier). Inject a vanguard fraction f of agents pinned to the truth who keep running the discriminating experiment, and vary (a) f , (b) the trust topology, (c) the vanguard’s *placement*, and (d) the rescue strength. We measure whether and how fast the mainstream flips.

4.3 Manipulated variables

Parameter	Implementation	Levels
vanguard fraction f	both	{0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50}
trust topology <code>kind</code>	both	{Watts–Strogatz, ring lattice, random (ER), scale-free (BA)}
vanguard placement	both	{random, high-centrality, clustered-behind-wide-bridge}
<code>mean_degree</code>	both	{2, 4, 8}
<code>social_weight</code> β_s	(S)	{0, 0.5, 1.0, 2.0}
<code>q_reliability</code>	(P)	{0.6, 0.8, 0.95}

4.4 Fixed parameters

Mainstream sealed at `cost_scale=1.0` / $g=100$ (above the Experiment-2 frontier); $N=120$ (larger, to resolve f in 5% steps); $\theta^*=1$; horizon =300; $n_{\text{seeds}}=50$; `gamma_policy=5` (finite, so escape is possible but not trivial).

4.5 Outcomes and analysis

Primary outcome: terminal mainstream conversion `occ_B` (excluding the pinned vanguard) as a function of f , per (topology, placement) cell. Fit each curve for a critical fraction f^* (the inflection of a sigmoid) and a sharpness; $f^*=undefined/linear \Rightarrow$ simple contagion. Secondary: conversion time; the order parameter m_t trajectory; in (S), the time-course of `rho_k` on the disagreement channels (does the seal physically re-open?).

Hypothesis 5 (Simple contagion). *Mainstream conversion is monotone and approximately linear in f with no threshold; topology and placement have only second-order effects.*

Hypothesis 6 (Complex contagion). *There is a sharp critical fraction f^* ; f^* depends on topology and placement—lower for clustered-behind-wide-bridge than for random or high-centrality placement—and rises with the depth of the self-seal (`cost_scale`, g) and falls with β_s and `q_reliability`.*

Prediction 3. *We expect complex contagion (Hypothesis ??): a threshold near $f^* \sim 0.2\text{--}0.25$ for random placement, materially lower for clustered wide-bridge placement, and a non-monotone (over-shooting, hence discounted) effect for an isolated high-centrality vanguard, consistent with the notebook-10 isolated-vanguard overconfidence result. Higher β_s and `q_reliability` shift f^* down.*

Severe test / falsifier. The deafened-vanguard control: set the mainstream’s trust toward vanguard agents to zero (zero the corresponding columns of T , renormalise). Social rescue must then vanish *at every f* . If the mainstream still converts, the “rescue” is leaking through an unmodelled channel and the trust-bandwidth-limited claim is refuted. A second falsifier: if conversion is linear in f with no topology dependence, paradigm change in this model is a *simple* contagion and the complex-contagion framing is wrong.

5 Shared infrastructure and reporting standards

Order parameters (existing code). (S): `Network.run_trace` $\rightarrow (m_t, \text{grav_attn}_t)$; `run_trace_bmr` $\rightarrow (m_t, \Delta F_t)$; `run_trace_precision` $\rightarrow (m_t, \text{gw}_t, \rho_t)$. (P): `run_fast` $\rightarrow \{\text{mean_qB}, \text{occ_B}, \text{p_discrim}, \text{chosen_d}, \text{final_q}\}$. No new instrumentation is required; Experiments 1–3 are sweeps over these existing returns.

Statistics. $n_{\text{seeds}} \geq 40$ per cell; report median and inter-quartile range of every order parameter, not just the mean (lock-in is heavy-tailed across seeds). Phase boundaries are reported with bootstrap confidence bands over seeds. All hypothesis tests are pre-registered one-sided tests against the stated null.

Multiple realisability as a corroboration criterion. Friston’s discipline: every mechanism claim must appear in *both* implementations or be flagged as an artifact. Self-censorship (Exp. 2) must show as a `chosen_d` collapse in (P) *and* a `rho_k` silencing in (S); social rescue (Exp. 3) must show as `occ_B` conversion in (P) *and* a re-opening of `rho_k` in (S).

No silent caps. Any truncation (max horizon, seed count, swept range) that bounds a reported boundary must be logged next to the result, so a frontier that is really an artifact of a too-short horizon cannot be mistaken for a structural limit.

6 Summary: philosophy $\rightarrow$ knob $\rightarrow$ prediction $\rightarrow$ falsifier

Construct	Knob(s)	Prediction	Falsifier
Kuhnian irreversibility vs. relaxation (Exp. 1)	<code>core_governance</code> , <code>gamma_policy</code> , ramp rate r	$A_{\text{hys}} \propto r$; zero residual loop at finite g ; fold only at $\gamma=\infty$	Residual loop $A_{\text{hys}}(r \rightarrow 0) > 0$ at any finite g
Progressive vs. degenerating programme (Exp. 2)	<code>cost_scale</code> , D_0 , <code>gamma_policy</code> , <code>experiment_bias</code>	<code>chosen_d</code> collapses above conviction ~ 0.7 even at zero cost (greedy)	At <code>cost=0</code> , <code>chosen_d</code> never collapses for any D_0
Simple vs. complex contagion / committed minority (Exp. 3)	f , topology kind, placement, β_s , <code>q_reliability</code>	Sharp $f^* \sim 0.2\text{--}0.25$, topology-dependent; isolated central vanguard over-shoots	Linear, topology-free conversion; or rescue survives deafened-vanguard control

Epistemic status. This is a Stage-1/2 design document (forest-walk synthesis): the experiments and their falsifiers are specified, but the predicted magnitudes are extrapolations from notebooks 18–20 and from the cited literature, not yet established on this exact protocol. The next step is to implement the three sweeps as notebooks 21–23 and run the falsifiers *first*.

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